INDIVIDUAL PROJECT REPORT

ON

“Arya College ClubSphere: Centralized Club and Event Management Platform”

INDIVIDUAL CONTRIBUTION REPORT — DASHBOARDS, TASKS & NOTIFICATIONS

Submitted in partial fulfillment of the requirement for the award of Degree of

Bachelor of Technology in Computer Science & Engineering

Submitted to:

Rajasthan Technical University, Kota (Raj.)

Submitted by

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Individual responsibility: Dashboards, Tasks & Notifications

PBL Team ID: PBL2627-CSE-A-024

Under the guidance of

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Department of Computer Science & Engineering

Session: 2026–27

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CERTIFICATE

This is to certify that the individual project report entitled “Arya College ClubSphere — Dashboards, Tasks & Notifications” has been prepared by Aditi Maheshwari (24EARCS007), a student of Bachelor of Technology in Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, affiliated with Rajasthan Technical University, Kota, during the academic session 2026–27.

The report records the student’s individual contribution to the group project Arya College ClubSphere (PBL2627-CSE-A-024), carried out under the guidance of Er. Ram Babu Buri. The contribution described here forms part of a four-member team project; the work of the other team members is acknowledged and is not claimed in this report.

Role	Name	Signature
Project Guide	Er. Ram Babu Buri Dept. of CSE, ACEIT, Jaipur	
Head of Department	Dr. Akhil Pandey Dept. of CSE, ACEIT, Jaipur	
Student	Aditi Maheshwari (24EARCS007)	

Date: _____ Place: Jaipur

DECLARATION

I, Aditi Maheshwari (24EARCS007), a student of B.Tech. Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, hereby declare that this individual report describes my own contribution to the group project Arya College ClubSphere, carried out during the session 2026–27 under the guidance of Er. Ram Babu Buri.

The technical content of this report was prepared from the project source code, database schema and module documentation. Features built by teammates are stated as such. The report does not claim sole authorship of ClubSphere or measured production performance.

To the best of my knowledge, this report has not been submitted to any other university or institution for the award of any degree or diploma.

Aditi Maheshwari

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B.Tech. (CSE), Session 2026–27

Signature: _____ Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, Er. Ram Babu Buri, for his guidance throughout the planning and development of ClubSphere. His emphasis on clear requirements and honest reporting of status shaped the way I designed and documented the Dashboards, Tasks & Notifications module.

I am thankful to Dr. Akhil Pandey, Head of the Department of Computer Science & Engineering, Arya College of Engineering & I.T., for the academic environment, and to the faculty of the department for their support.

I am grateful to my teammates — Avantika Mishra, Avanshi Kothari and Akansha Sharma. My module depends on a shared MySQL schema, AuthenticationFilter and JSP chrome. Working with Git taught me integration and communication within a team.

Finally, I thank my family and friends for their encouragement and patience during the project.

Aditi Maheshwari

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ABSTRACT

This report documents my contribution to ClubSphere: Dashboards, Tasks and Notifications. I implemented role-specific dashboards for Admin, Club Head, Club Member and Student, club task assignment and status updates, announcements, in-app notifications with unread counts, and report summaries. Shared navbar and sidebar chrome also sit in this module so that every role sees a consistent workspace after login.

Keywords: Dashboard, tasks, notifications, JSP, ReportService

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CHAPTER 1 — INTRODUCTION

1.1 Background

ClubSphere is the group PBL project (PBL2627-CSE-A-024) that centralises clubs, events, QR registration, tasks and notifications for Arya College of Engineering & I.T. The team report describes the whole system. This individual report documents only my contribution: **Dashboards, Tasks & Notifications**.

1.2 Purpose of this Report

This report separates my work from that of my teammates, explains the design of my module, how it was tested, and what remains incomplete.

1.3 Problem Addressed by My Module

After login, users needed a home that matches their role. Club work (tasks) and notices had to appear inside the same portal rather than in a chat group.

1.4 Objectives of My Contribution

- Route each role to a useful dashboard with counts and lists.
- Let club heads assign tasks and publish announcements.
- Let members update task status.
- Provide an in-app notification inbox and unread count.

CHAPTER 2 — PROJECT OVERVIEW AND INDIVIDUAL RESPONSIBILITIES

2.1 Overall Project Context

ClubSphere is developed by four students and hosted at github.com/Avantikamishra862/Arya_Clubsphere. It is a Java 17 / JSP / Servlet / MySQL application deployed as a Tomcat WAR.

2.2 Module Ownership

Member	Enrollment	Module
Avantika Mishra	24EARCS032	Authentication & User Management
Avanshi Kothari	24EARCS031	Club Management
Akansha Sharma	24EARCS012	Event Management & QR Code Registration
Aditi Maheshwari	24EARCS007	Dashboards, Tasks & Notifications

My owned area is highlighted in the implementation chapter. Shared files (schema.sql, AuthenticationFilter, navbar.jsp, pom.xml) were coordinated with the team.

2.3 In-scope and Out-of-scope

In scope: Role dashboard JSPs, TaskService, NotificationService, ReportService, NotificationServlet, navbar/sidebar/footer, announcement views.

Out of scope: I do not claim BCrypt login internals or QR PNG generation except consuming session unreadCount set at login.

CHAPTER 3 — TECHNOLOGY STACK

I used the shared ClubSphere stack. My module specifically depends on the items below.

Layer	Technology	How I used it
UI	JSP dashboards, Bootstrap, sidebar	Role workspace
Data	tasks, announcements, notifications, activity_logs	Club work and inbox
Web	ClubHeadServlet task/announcement paths, ClubMemberServlet, NotificationServlet, ReportService	Assign, update, inbox, charts/counts

CHAPTER 4 — REQUIREMENTS OF MY MODULE

4.1 Users

All authenticated roles land on a dashboard. Club heads assign work. Members execute tasks. Everyone has an inbox.

4.2 Functional Requirements

ID	Requirement
FR-D1	Role dashboard after login
FR-D2	Assign and update tasks
FR-D3	Publish announcements (ALL / MEMBERS_ONLY)
FR-D4	Notification list, mark one/all read
FR-D5	Admin and club-head report views

4.3 Non-functional requirements

Sidebar links must match AuthenticationFilter. Dashboard queries should avoid obvious N+1 patterns. Unread counts should refresh after mark-read.

CHAPTER 5 — DESIGN AND IMPLEMENTATION

5.1 Request flow

Login stores unreadCount. Dashboards call ReportService plus list services. Task create notifies the assignee. NotificationServlet GET lists; POST marks one read as JSON.

5.2 Principal classes and views

Layer	Items
Views	admin/club-head/club-member/student dashboard JSPs, tasks, announcements, reports, notifications.jsp
Services	TaskService, NotificationService, ReportService
Chrome	navbar.jsp, sidebar.jsp, footer.jsp

5.3 Implementation notes

Admin dashboard shows overall stats. Club-head dashboard is club-scoped. Members see assigned tasks. Students see registrations and profile shortcuts. Activity logs record login.

UML figures for the whole system are in the team report and in docs/UML/Arya_College_ClubSphere_UML_Diagrams.pdf.

CHAPTER 6 — TESTING AND VERIFICATION

6.1 Strategy

Manual: login as each seeded role and confirm landing URL; assign a task as head_coding and complete it as member_coding_1; publish announcement; mark notifications read.

6.2 Representative checks

Check	Result / note
Admin landing	/admin/dashboard
Club head landing	/club-head/dashboard
Task status update	Member can complete assigned task
Mark all read	unreadCount becomes 0

CHAPTER 7 — CHALLENGES AND LEARNING OUTCOMES

7.1 Challenges

Keeping sidebar links aligned with the filter; aggregating dashboard counts without heavy queries; keeping session unreadCount consistent after new inserts.

7.2 Learning outcomes

I learned that a dashboard is an integration surface: it reads every module’s tables and must fail safely when a club head has no club assigned.

CHAPTER 8 — CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Dashboards turn the other modules into a daily workspace. Tasks and notifications close the loop after an event is published.

8.2 Future work in my module

Email/push notifications, richer charts, and automated UI tests for each role home.

REFERENCES

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